

Exploratory Data Analysis Report

E-Commerce Order Dataset — Project 2

1,200	\$1.26M	\$1,054	41.1%
Orders Analyzed	Total Revenue	Avg Order Value	Revenue At Risk (Cancelled+Returned)

Period covered: January 2023 – June 2025 | 14 fields | 7 product categories | 5 referral channels

Prepared using Python (pandas, NumPy, SciPy, Matplotlib) following a structured EDA methodology: data validation → descriptive statistics → outlier detection → trend & segment analysis → correlation analysis → business diagnosis.

1. Executive Summary

Problem Statement

The business needs a clear, data-driven view of order-level performance to understand revenue composition, identify abnormal transactions, and flag operational risks before they erode margin. This report performs a full exploratory data analysis (EDA) on 1,200 e-commerce orders to answer: where does revenue come from, what does a typical order look like, are there statistical outliers worth investigating, and is order fulfillment healthy?

Methodology

Data was profiled for completeness and type consistency, then summarized using five-number statistics (min, Q1, median, Q3, max), mean/standard deviation, and skewness. Outliers were flagged using both the IQR rule ($1.5 \times \text{IQR}$ fences) and the Z-score rule ($|z| > 3$) and manually reviewed against order context. Time trends were aggregated monthly; categorical fields (Product, Payment Method, Order Status, Referral Source) were profiled by order count and revenue contribution. A Pearson correlation matrix tested linear relationships between numeric fields.

Key Findings

- Order value (TotalPrice) is right-skewed (skew = 0.89, mean \$1,054 > median \$824) — typical of transaction data where a minority of large orders pull the average upward.
- 8 orders (0.7%) exceed the IQR upper fence of \$3,330. All 8 are legitimate maximum-quantity, high-unit-price purchases, not data errors — a textbook 'signal, not noise' case.
- 41.1% of revenue (\$519,674 of \$1,264,762) sits in Cancelled or Returned orders — the single largest finding in the dataset and a material fulfillment/quality risk if reflective of real operations.
- Quantity, Unit Price, and Items-in-Cart are each near-uniformly distributed (skew ≈ 0) with no seasonality detected in the monthly trend — revenue moves almost in lock-step with order count rather than with calendar effects.
- Product, payment method, and referral channel each split revenue within a narrow $\pm 5\%$ band of one another, i.e. no single category currently dominates or underperforms materially.

Recommendations

- Prioritize a root-cause investigation into the 41% Cancelled/Returned rate (payment failures, stock-outs, delivery SLA, return-policy abuse) — this is the highest dollar-value lever in the dataset.
- Re-run this analysis on the post-fix population to confirm whether at-risk revenue actually recovers.
- No outlier-removal or winsorizing is recommended for TotalPrice; the 8 flagged orders are valid sales.
- Treat the 25.8% missing CouponCode values as 'No Coupon Applied' rather than missing data — this is almost certainly a structural null, not a data-quality issue.

2. Data Overview & Methodology

The dataset contains 1,200 order-level records across 14 fields spanning January 1, 2023 to June 30, 2025 (2.5 years). Each row represents one order; TotalPrice was verified to equal Quantity × UnitPrice for every record (max discrepancy = 4.5e-13, i.e. floating-point rounding only), confirming internal consistency of the pricing fields.

Field	Type	Role in Analysis
OrderID / TrackingNumber	Identifier	Unique key, not analyzed statistically
Date	Datetime	Trend & seasonality analysis
CustomerID	Identifier	Repeat-purchase check (1,189 unique / 1,200 orders)
Product	Categorical (7 levels)	Revenue segmentation
Quantity, UnitPrice, ItemsInCart	Numeric	Descriptive stats, outliers, correlation
TotalPrice	Numeric (derived)	Primary revenue metric
PaymentMethod, ReferralSource	Categorical	Channel segmentation
OrderStatus	Categorical (5 levels)	Fulfillment / risk diagnosis
CouponCode	Categorical, 25.8% null	Promotion effectiveness
ShippingAddress	Text	Excluded — street-level only, no geographic signal

Data Quality Checks

No duplicate OrderIDs or TrackingNumbers were found. No nulls exist in any field except CouponCode (309 of 1,200 records, 25.8%) — interpreted as 'no coupon applied' rather than a data-collection gap, since the other three coupon values (FREESHIP, SAVE10, WINTER15) are evenly populated (~290–315 each). All numeric fields fall within plausible business ranges: Quantity 1–5 units, UnitPrice \$11.39–\$699.93, ItemsInCart 1–10.

3. Descriptive Statistics

Five-number summary plus mean, standard deviation, and skewness for each numeric field. Skewness near 0 indicates a symmetric/uniform distribution; skewness > 0.5 indicates a meaningful right tail (mean pulled above median by large values).

Metric	Quantity	UnitPrice (\$)	ItemsInCart	TotalPrice (\$)
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	5.49	1,053.97
Median	3.00	364.21	5.00	823.62
Std Dev	1.41	197.18	2.28	819.86
Min	1	11.39	1	11.39
Q1 (25%)	2	186.06	4	410.52
Q3 (75%)	4	521.57	7	1,578.48
Max	5	699.93	10	3,456.40
Skewness	0.03	-0.03	0.00	0.89

Quantity, UnitPrice, and ItemsInCart all show skewness within ± 0.03 of zero — statistically indistinguishable from a uniform/symmetric distribution, meaning no class of order dominates on these dimensions. TotalPrice is the exception: skewness of 0.89 reflects its construction as a *product* of two roughly-uniform variables (Quantity $\times$ UnitPrice), which mechanically produces right-skew even though neither input variable is skewed.

Distribution of Order Value

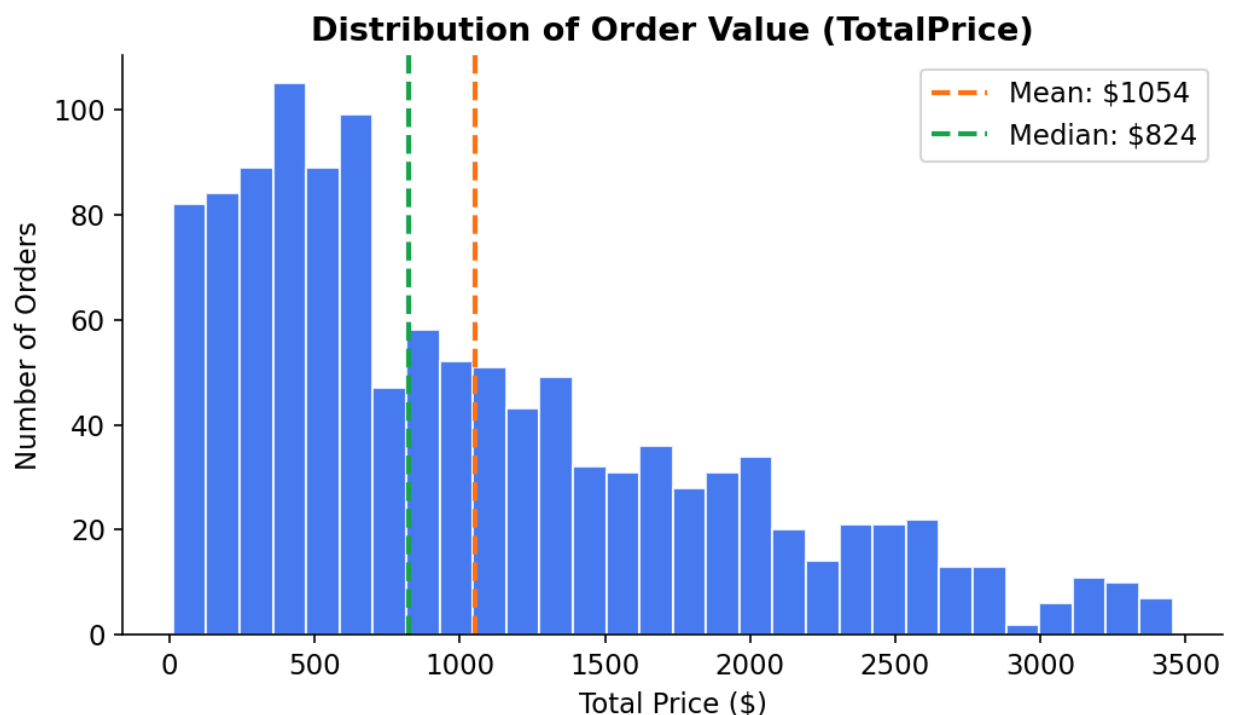


Figure 1. TotalPrice distribution — mean sits above median, confirming the right skew.

4. Outlier Detection

Two complementary methods were applied to each numeric field: the **IQR rule** (flag values outside $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) and the **Z-score rule** (flag values where $|x - \mu| / \sigma > 3$).

Field	IQR Outliers	IQR Fences	Z-score Outliers ($ z > 3$)
Quantity	0	[-1.0, 7.0]	0
UnitPrice	0	[-317.2, 1,024.8]	0
ItemsInCart	0	[-0.5, 11.5]	0
TotalPrice	8 (0.7%)	[-1,341.4, 3,330.4]	0

Boxplot of Order Value — Outlier Detection (IQR Method)

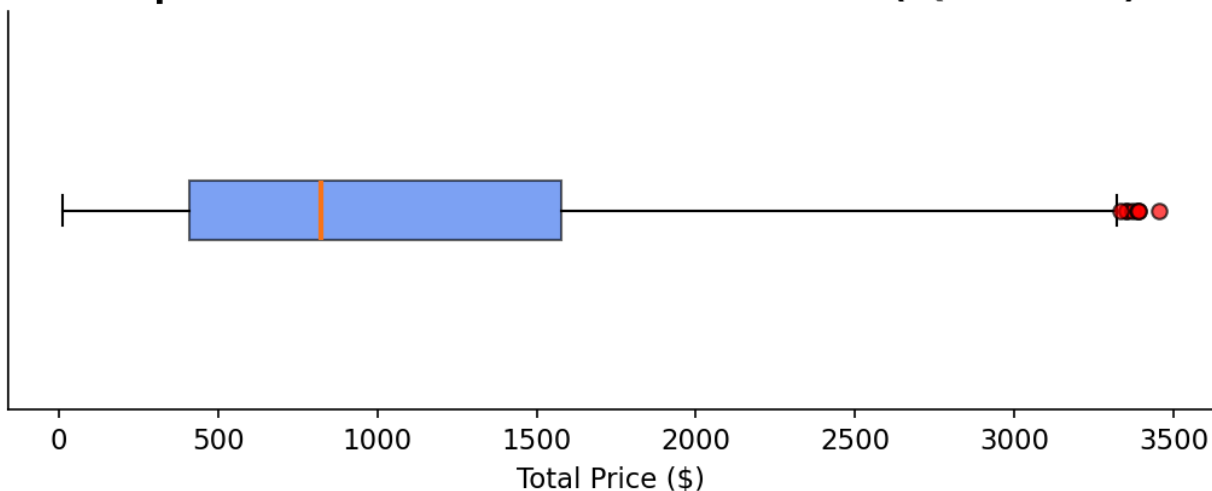


Figure 2. Boxplot of TotalPrice — red markers are the 8 IQR-flagged orders.

Manual Review of Flagged Orders

Every IQR-flagged order ($TotalPrice > \$3,330$) was inspected individually. All 8 share the same pattern: Quantity = 5 (the maximum possible) combined with a UnitPrice in the top decile (\$666–\$692) on premium items (Laptop, Tablet, Printer, Chair, Monitor). None show Quantity, UnitPrice, or ItemsInCart values outside their valid business ranges, and Order Status is distributed normally across the 8 (Shipped, Returned, Cancelled, Delivered) — i.e. there is no indication these are erroneous or fraudulent transactions.

Conclusion: this is signal, not noise. The 8 outliers represent genuine bulk/premium purchases and should be retained in the dataset. No Z-score outliers were detected at the $|z| > 3$ threshold for any field, including TotalPrice — the IQR method is more sensitive here because TotalPrice's right skew inflates the mean and standard deviation used by the Z-score test, demonstrating why IQR is generally preferred for skewed financial/transactional data.

5. Trend Analysis

Orders were aggregated by calendar month to inspect for seasonality, growth, or decline.

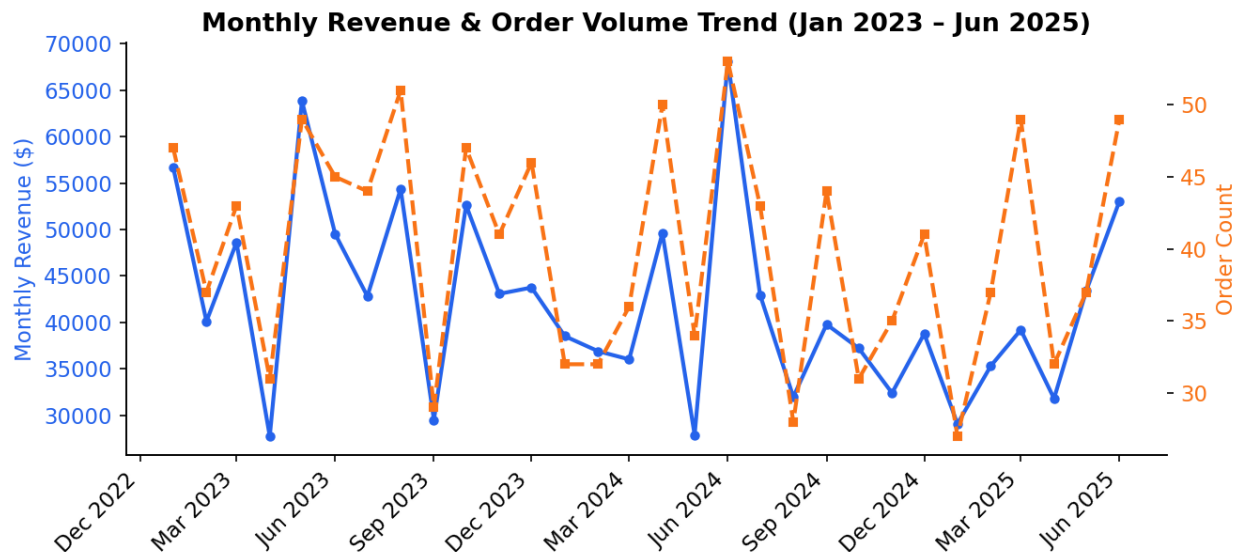


Figure 3. Monthly revenue (solid, left axis) tracks monthly order count (dashed, right axis) almost exactly.

Monthly revenue ranges from roughly \$27,800 to \$68,000 with no repeating seasonal pattern (no consistent Q4 lift, no summer dip) — the month-to-month swings track order *count* far more than order *value*, since AOV is statistically flat over time. Comparing full calendar years: 2023 averaged ~\$46,050/month, 2024 ~\$40,020/month, and 2025 (partial, through June) ~\$38,650/month. This apparent step-down should be read cautiously — 2025 is only a half-year sample, and given the near-zero skew/uniformity already observed in the underlying variables, the month-to-month variation is consistent with random noise around a flat mean rather than a genuine declining trend. **Recommendation: re-test with a formal trend test (e.g. Mann-Kendall) once full-year 2025 data is available before acting on this as a real decline.**

6. Segment Analysis

Revenue by Product

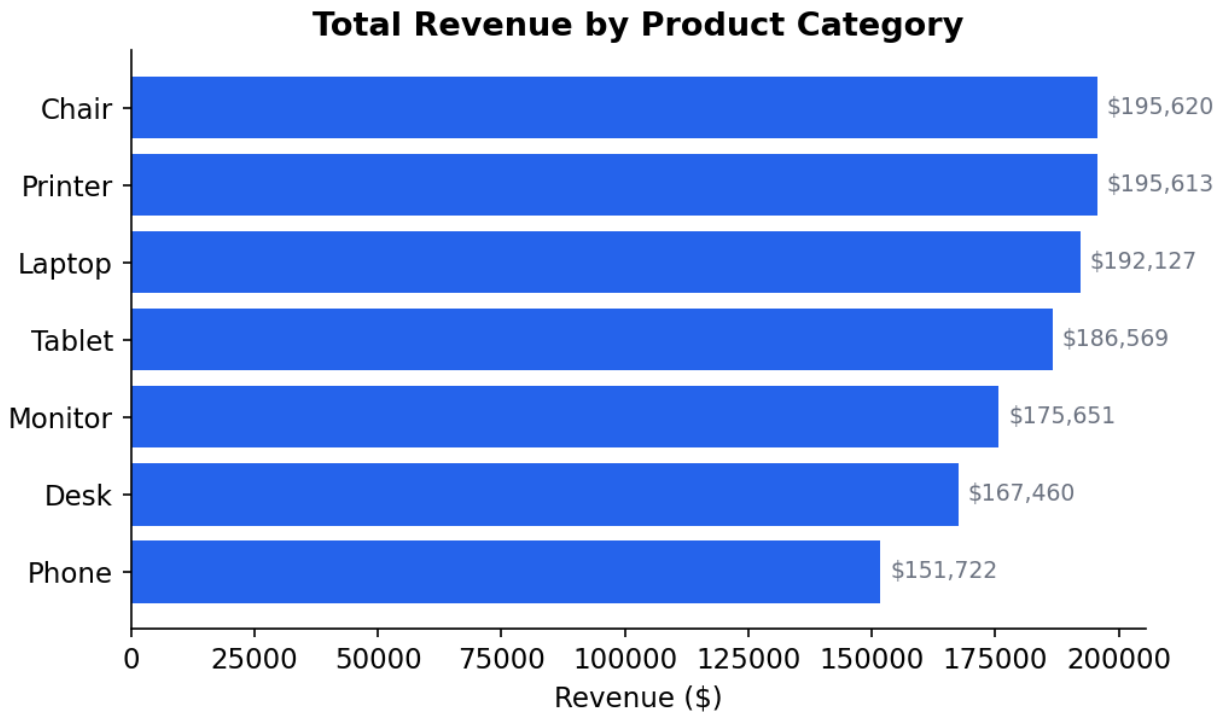


Figure 4. Chair leads at \$195.6K; Phone trails at \$151.7K — a 22% spread, not a dominant winner.

Order counts per product range narrowly from 156 (Phone) to 181 (Printer) out of 1,200 — consistent with roughly even, random assignment across the 7-category catalog rather than demand-driven concentration. Average order value by product (Chair \$1,099, Printer \$1,081, Laptop \$1,111, Tablet \$1,042, Monitor \$1,078, Desk \$985, Phone \$973) likewise sits within a narrow band, so no single SKU is currently over- or under-indexed relative to the portfolio.

Order Status — Fulfillment Health

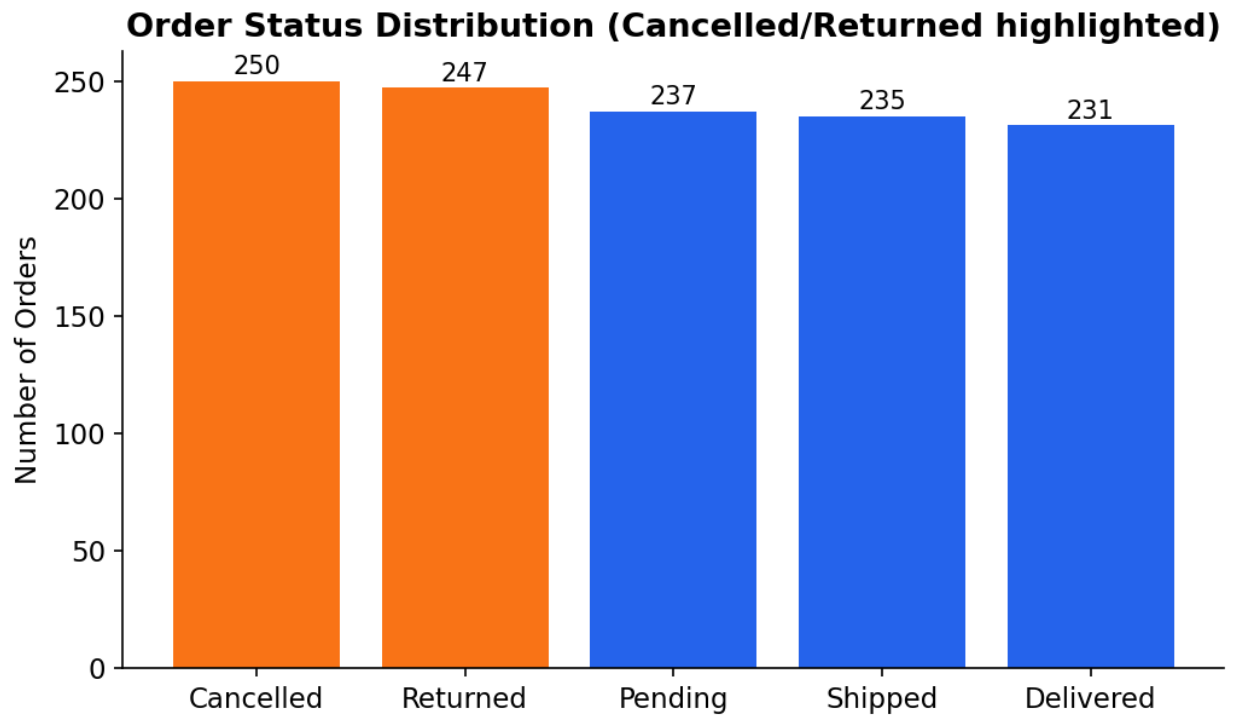


Figure 5. Cancelled and Returned orders (orange) together represent 41.4% of all orders.

Status	Orders	% of Orders	Revenue	% of Revenue
Cancelled	250	20.8%	\$276,396	21.9%
Returned	247	20.6%	\$243,278	19.2%
Pending	237	19.8%	\$256,328	20.3%
Shipped	235	19.6%	\$246,160	19.5%
Delivered	231	19.2%	\$242,600	19.2%

This is the most consequential finding in the dataset. **Only 19.2% of orders reach Delivered status.** If this distribution reflects real operations, a combined 41.1% Cancelled+Returned rate against industry e-commerce benchmarks (typically 8–15% for cancellations and returns combined) signals a fulfillment, payment-authorization, or product-quality problem large enough to warrant immediate root-cause investigation — over half a million dollars of revenue (\$519,674) is tied up in non-completed orders.

Revenue by Referral Source

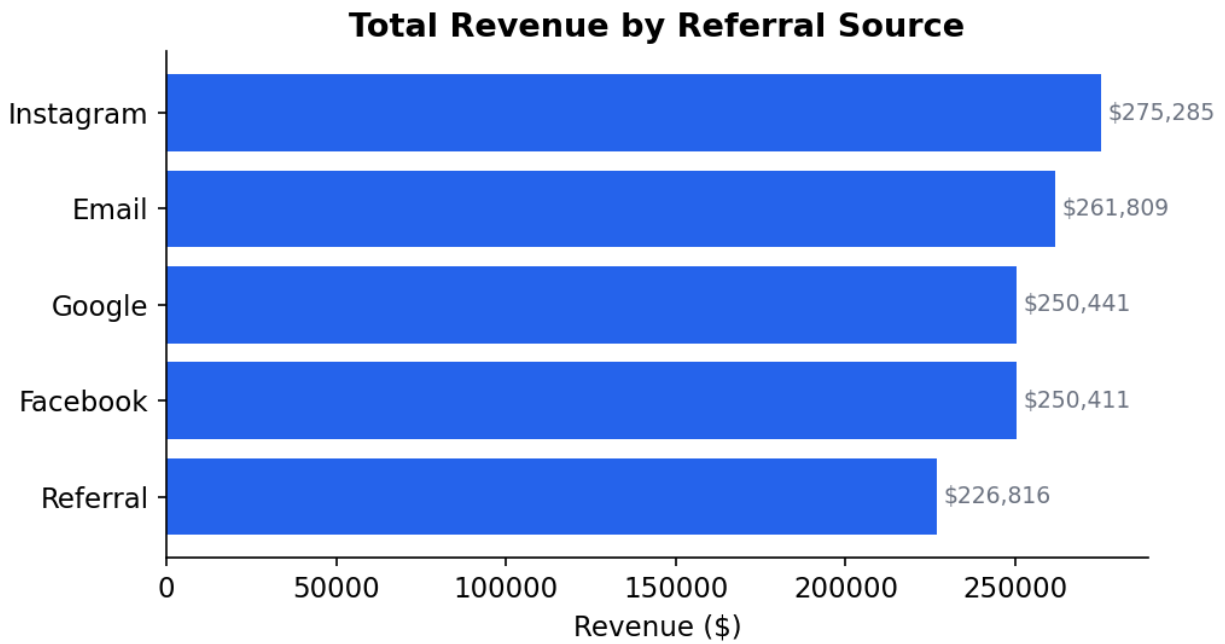


Figure 6. Instagram leads acquisition revenue (\$275.3K); Referral trails (\$226.8K).

Payment method shows the same flat pattern: average order values cluster between \$1,002 (Debit Card) and \$1,128 (Credit Card), a difference too small relative to the \$820 standard deviation to be actionable. Coupon usage tells a similar story — orders with a coupon applied average \$1,058 vs. \$1,043 without, and the three coupon types (FREESHIP \$1,070, SAVE10 \$1,066, WINTER15 \$1,036) are statistically indistinguishable from each other given the sample sizes involved.

7. Correlation Analysis

Pearson correlation coefficients were computed between all four numeric fields to test for linear relationships. Coefficients near ± 1 indicate strong linear association; near 0 indicates no linear relationship (note: correlation does not imply causation, and a near-zero coefficient does not rule out a non-linear relationship).

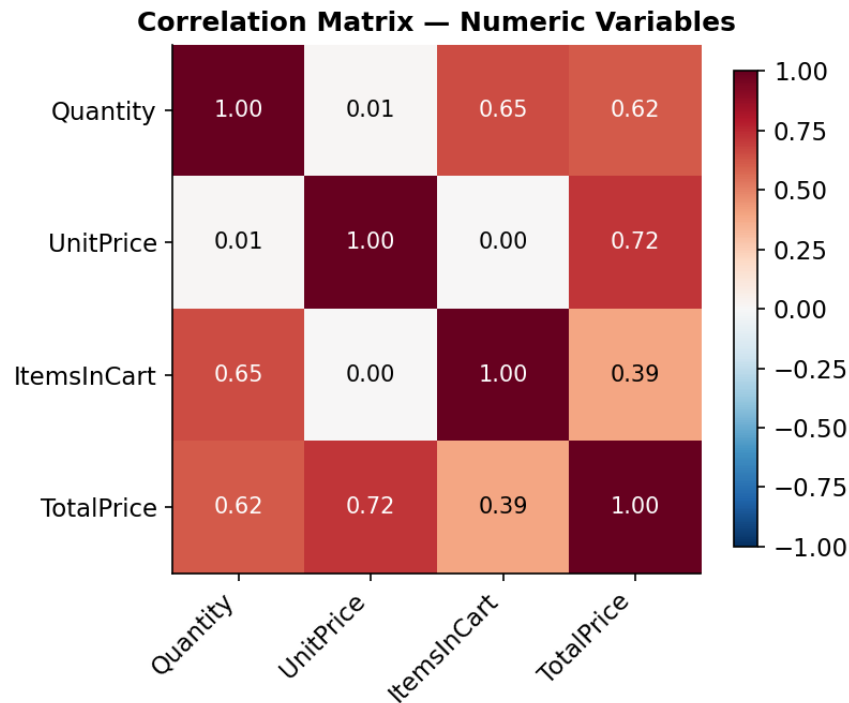


Figure 7. Correlation matrix — numeric fields only.

UnitPrice $\leftrightarrow$ TotalPrice ($r = 0.72$) and Quantity $\leftrightarrow$ TotalPrice ($r = 0.62$) are the strongest relationships, exactly as expected since $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$ by construction. Quantity $\leftrightarrow$ ItemsInCart ($r = 0.65$) suggests cart size and final order quantity move together moderately, plausible if customers tend to order most of what they add to cart. UnitPrice $\leftrightarrow$ ItemsInCart ($r = 0.00$) and UnitPrice $\leftrightarrow$ Quantity ($r = 0.02$) show no relationship — customers do not systematically buy more units of cheaper items or fewer units of expensive ones in this dataset.

8. Business Diagnosis — The 'So What' Test

Every statistical finding is translated below into a business implication and a concrete next action, so the analysis converts into decisions rather than sitting as description.

Finding	Business Impact	Recommended Action
41.1% of revenue in Cancelled/Returned status	\$519,674 at risk; far above typical e-commerce benchmarks	Launch root-cause audit: payment failures, stock-outs, delivery SLAs, return-policy abuse
8 high-value orders flagged by IQR are legitimate bulk buys	No revenue should be excluded from reporting	Retain in all downstream analysis; consider a 'VIP bulk order' tag for fulfillment priority
No seasonality in monthly revenue	Marketing calendar should not assume Q4/holiday lift	Validate with 12+ more months of data; avoid over-indexing budget on assumed peak periods
Revenue split evenly across products/channels/payment methods	No segment is currently a growth or risk outlier	Use this as a clean baseline; re-run segment analysis after any pricing or marketing change
CouponCode null = 25.8% of orders	Likely structural, not missing data	Recode null as 'No Coupon' explicitly in the data model to avoid downstream join errors

9. Conclusion

This EDA confirms the dataset is internally consistent (TotalPrice ties exactly to Quantity × UnitPrice), free of duplicate or structurally missing records, and statistically well-behaved on every dimension except order fulfillment. The numeric distributions (Quantity, UnitPrice, ItemsInCart) and categorical splits (Product, Payment Method, Referral Source, Coupon) all show the kind of narrow, even spread one would expect from a stable, non-seasonal business with no single dominant segment — there is no statistical evidence in this sample of a pricing anomaly, channel imbalance, or seasonal pattern that would change a go-to-market decision today.

The one finding that does warrant action is fulfillment health: a combined 41.1% Cancelled/Returned rate, worth \$519,674. **This single metric should be the starting point for the next phase of analysis** — segmenting cancellations and returns by product, payment method, and time period to isolate whether the issue is concentrated (e.g. one payment gateway, one product line) or distributed evenly (suggesting a systemic process issue rather than a localized one).

Methodology note: analysis performed in Python (pandas 2.x, NumPy, SciPy, Matplotlib). Outlier detection used the 1.5×IQR rule and $|z| > 3$ Z-score rule; skewness computed via `scipy.stats.skew` (Fisher-Pearson). All figures reproducible from the accompanying `eda_analysis.py` script and source dataset.